



Putting Our Heads Together



*THE IMPACT OF CONCUSSIONS ON PLAYER VALUE IN
PROFESSIONAL SPORTS.*

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If I am still battling the effects of my concussion a year later, what happens to professional athletes?

How does this invisible injury affect their overall performance and market value?

What is a Concussion?



A traumatically induced alteration in mental status, caused by a direct impact to the head or when the brain is subjected to rotational acceleration.

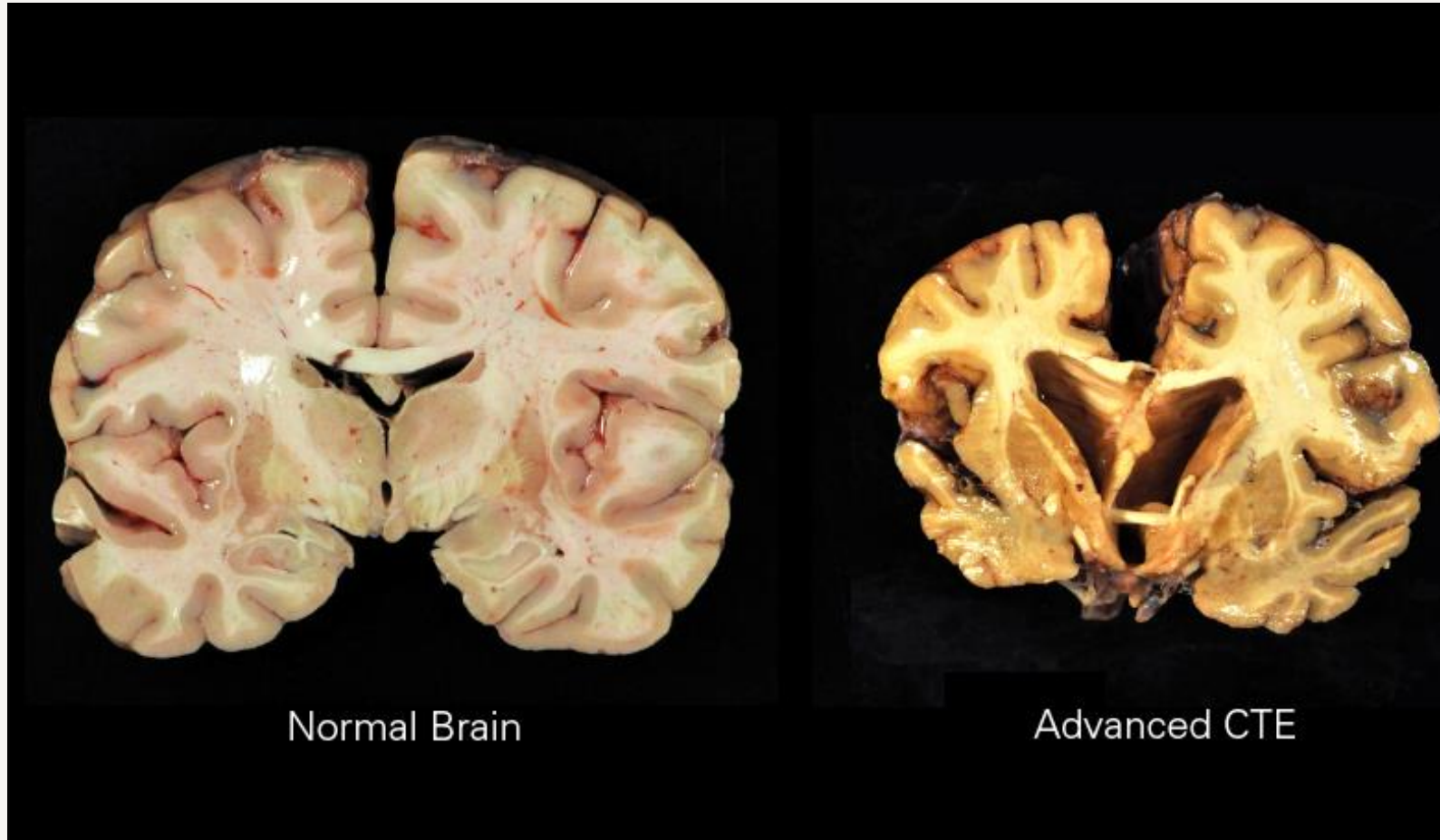
Grade	Cantu ^[11]	Colorado ^[21]	Roberts ^[40]	American Academy of Neurology ^[42]
0			'Bell ringer'; no LOC; no PTA	
1	No LOC; PTA <30 min	No LOC; confusion without amnesia	No LOC; PTA <30 min	No LOC; transient confusion; concussion symptoms or mental status abnormality resolve in <15 min
2	LOC <5 min; PTA >30 min and <24h	No LOC; confusion with amnesia	LOC <5 min; PTA >30 min and <24h	No LOC; transient confusion; concussion symptoms or mental status abnormality last >15 min
3	LOC >5 min or PTA >24h	LOC	LOC >5 min or PTA >24h	Any LOC, either brief or prolonged

LOC = loss of consciousness; **PTA** = post-traumatic amnesia.

Table V. Post-concussion symptoms (PCS) scale^[55]

PCS	Severity scale (0 = none; 6 = severe)						
Dizziness	0	1	2	3	4	5	6
Headache	0	1	2	3	4	5	6
Nausea	0	1	2	3	4	5	6
Vomiting	0	1	2	3	4	5	6
Balance problems	0	1	2	3	4	5	6
Trouble falling asleep	0	1	2	3	4	5	6
Sleeping more than usual	0	1	2	3	4	5	6
Drowsiness	0	1	2	3	4	5	6
Sensitivity to light	0	1	2	3	4	5	6
Sensitivity to noise	0	1	2	3	4	5	6
More emotional than usual	0	1	2	3	4	5	6
Irritability	0	1	2	3	4	5	6
Sadness	0	1	2	3	4	5	6
Nervousness	0	1	2	3	4	5	6
Numbness or tingling	0	1	2	3	4	5	6
Feeling slowed down	0	1	2	3	4	5	6
Feeling like 'in a fog'	0	1	2	3	4	5	6
Difficulty concentrating	0	1	2	3	4	5	6
Difficulty remembering	0	1	2	3	4	5	6
Other	0	1	2	3	4	5	6

Chronic Traumatic Encephalopathy



RETURN-TO-PARTICIPATION PROTOCOL



PHASE
1

SYMPTOM LIMITED ACTIVITY

The player is prescribed rest, limiting or, if necessary, avoiding activities (both physical and cognitive) which increase or aggravate symptoms. Under athletic training staff supervision, limited stretching and balance training can be introduced, progressing to light aerobic exercise, all as tolerated.

AEROBIC EXERCISE

Under direct oversight of the team's medical staff, the player should begin graduated cardiovascular exercise and may also engage in dynamic stretching and balance training. Neurocognitive and balance testing can be administered after completion of Phase Two and the results should be interpreted as back to baseline.

PHASE
2

PHASE
3

FOOTBALL-SPECIFIC EXERCISE

The player continues with supervised cardiovascular exercises that are increased and may mimic sport-specific activities, and supervised strength training is introduced. The player is allowed to practice with the team in sport-specific exercise for 30 minutes or less with ongoing and careful monitoring.

CLUB-BASED NON-CONTACT TRAINING DRILLS

The player continues cardiovascular, strength and balance training, team-based sports-specific exercise, and participates in non-contact football activities (e.g. throwing, catching, running, and other position-specific activities). Neurocognitive and balance testing should be completed no later than the end of Phase Four with the results interpreted as back to baseline.

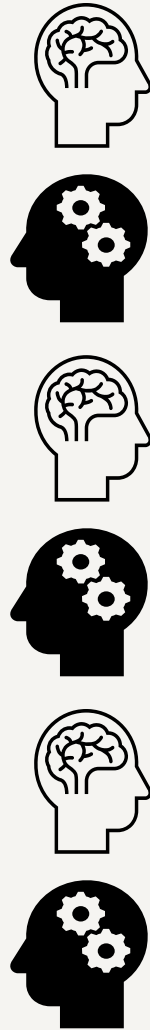
PHASE
4

PHASE
5

FULL FOOTBALL ACTIVITY / CLEARANCE

Upon clearance by the Club physician for full football activity involving contact, the player must be examined by the Independent Neurological Consultant (INC) assigned to his Club. If the INC concurs with the Club physician that the player's concussion has resolved, he may participate in his Club's next practice or game.

Revised July 2020



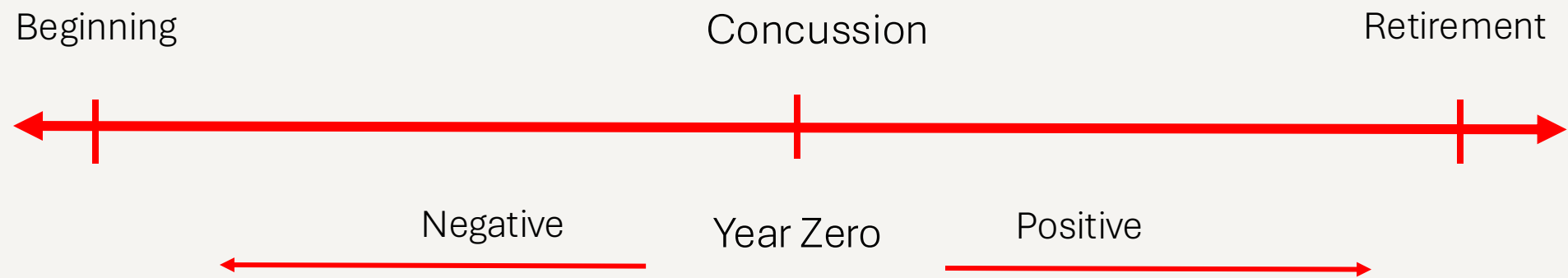
NHL paid \$70.6 million in legal fees
for concussion settlement that paid
players \$18.49 million



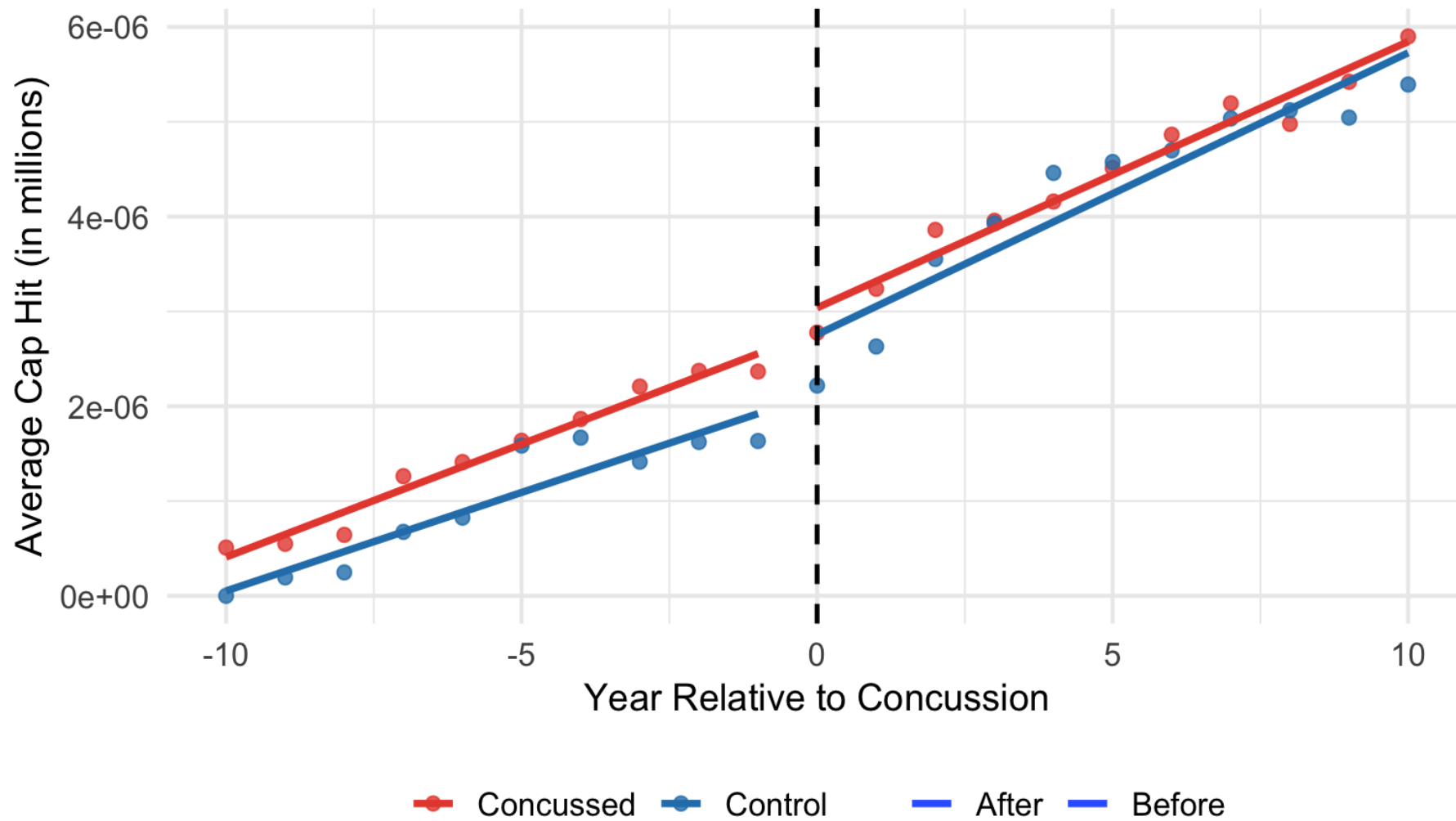
How does a concussion impact an NHL
player's market value?

Methodology

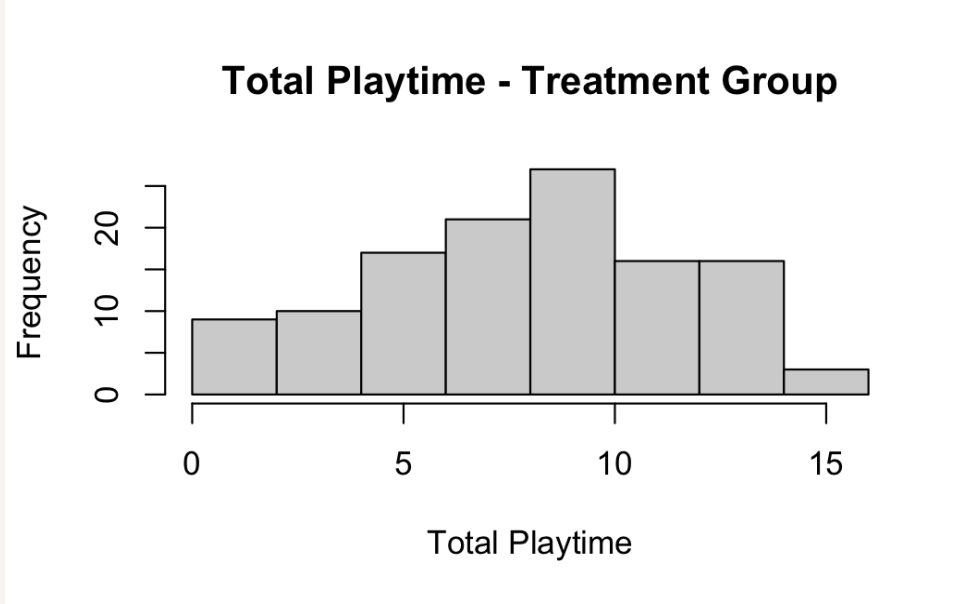
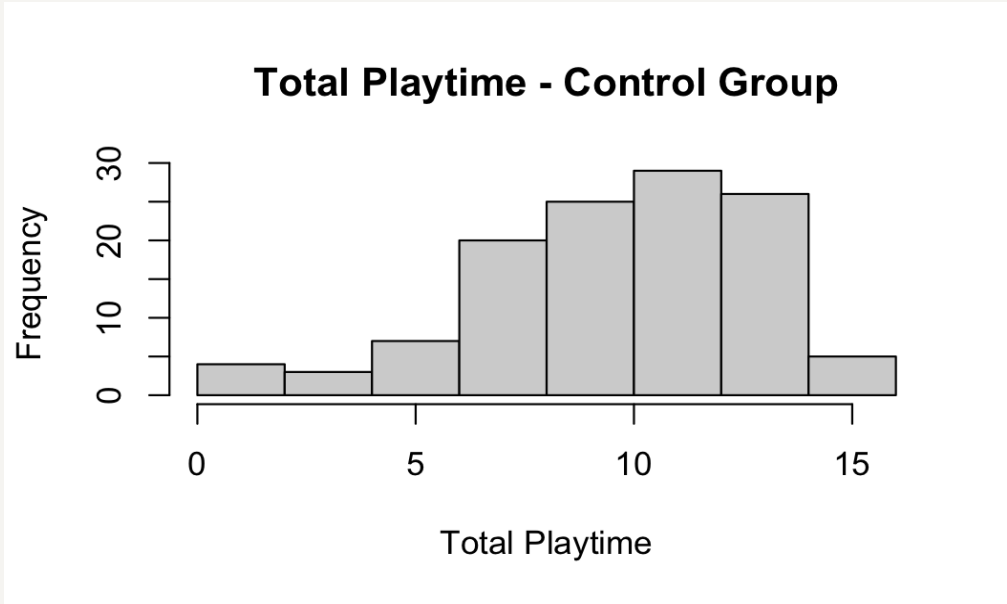
A Player's Career



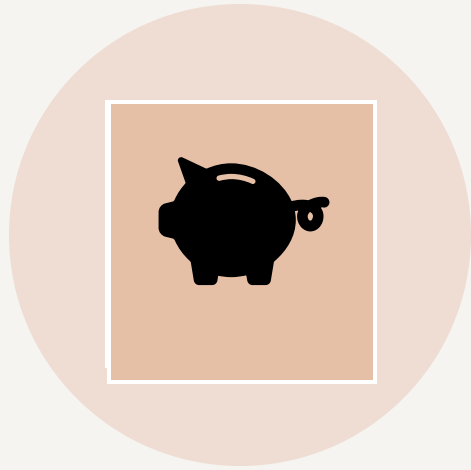
Player Cap Hit Before and After Concussion



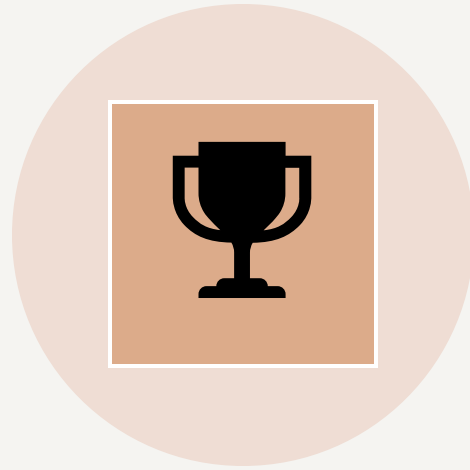
Summary Statistics by Group and Time Period									
Group	Period	Total GP	Total Points	Total Cap Hit (\$M)	Cap Hit / GP	Cap Hit / Point	Avg GP	Avg Points	
Concussed	After	49,791	24,276	3,910.750	0.079	0.161	54.1	26.4	★
Concussed	Before	27,647	12,800	995.687	0.036	0.078	53.7	24.9	
Control	After	84,055	42,068	6,024.045	0.072	0.143	59.0	29.5	★
Control	Before	18,792	9,268	533.376	0.028	0.058	52.3	25.8	



Why does this matter?



PLAYER EARNINGS



TEAM SUCCESS



COGNITIVE HEALTH

What should change?

- Strategic Planning
- Restructuring Contracts – Bonuses and Term
- Stanley Cup or Bust



STANLEY CUP[®]
PLAYOFFS



